Feedback suppression in 405 nm superluminescent diodes via engineered scattering
Supplementary Data

Andrea Martínez Pacheco[1], Antonio Consoli[1,2,§], Cefe López[1]
[1]Instituto de Ciencia de Materiales de Madrid (ICMM), Consejo Superior de Investigaciones Científicas (CSIC), Calle Sor Juana Inés de la Cruz, 3, 28049 Madrid, Spain.
[2]Escuela de Ingeniería de Fuenlabrada (EIF), Universidad Rey Juan Carlos (URJC), Camino del Molino 5, 28942 Fuenlabrada, Madrid, Spain.
§antonio.consoli@urjc.es

Section 1. Original Laser Diode.

In Fig. S1, we show the images of the original laser diode (LD) obtained after removing the TO-can encapsulation, with a TO-can opener (Thorlabs WR1).

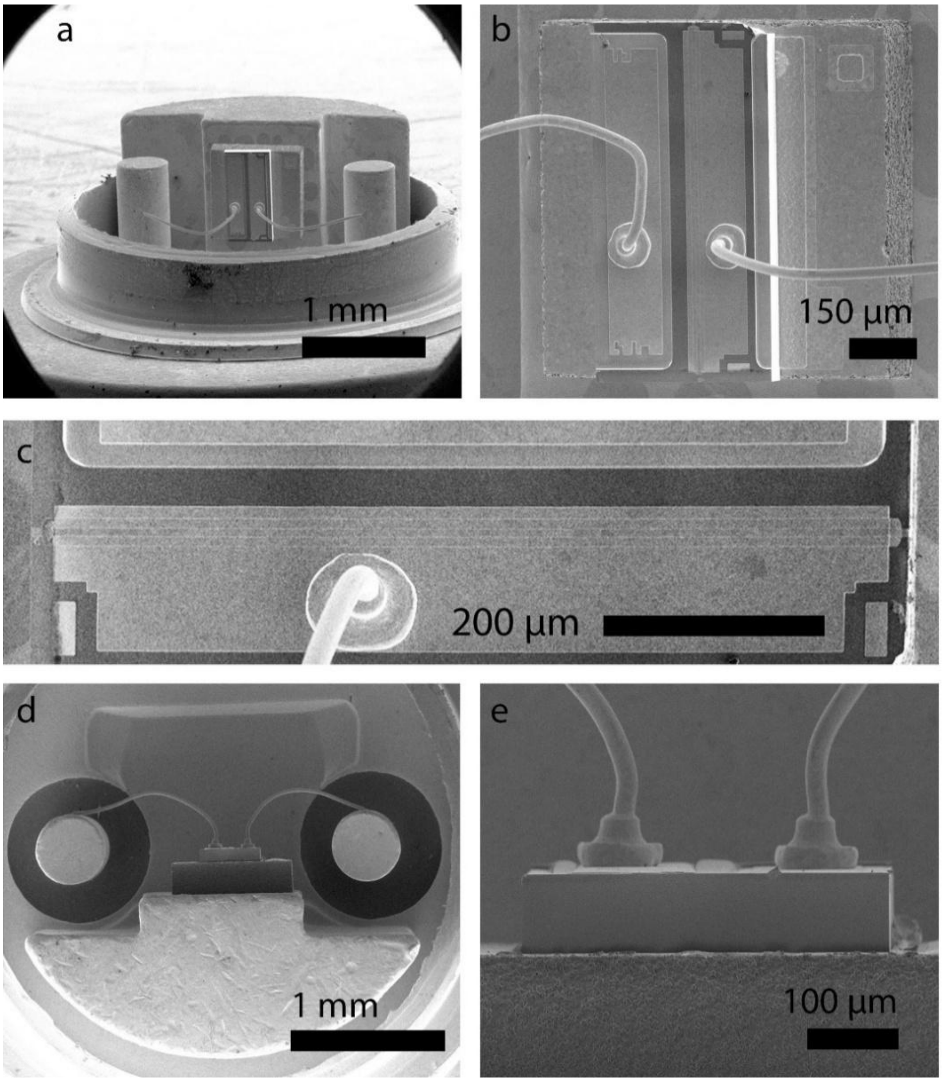

Figure S1. Scanning electron microscopy images of the original laser diode (LD) at different views and scales. Top views with scale bars of 1 mm (a), 150 µm (b), and 200 µm (c). Edge views with scale bars of 1 mm (d) and 100 µm (e).

Specifically, Fig. S1a shows a top view of the LD over a wide field of view, where the remaining circular metal rim of the TO-can package is visible, together with the anode and cathode terminals. Figure S1b shows a top-view image of the device over a smaller field of view, where only the laser die is visible. The anode (right) and cathode (left) gold pads can be identified. The laser cavity is located on the anode side, and a magnified view is provided in Figure S1c. The cavity consists of a ridge waveguide forming a Fabry–Pérot structure with a length of $L \approx 790$ μm. An edge view of the LD is also provided. In the large field-of-view image (Fig. S1d), the electrical terminals and part of the remaining circular TO-can are visible. The smaller-scale view (Fig. S1e) shows the substrate in greater detail.

The emission of the original LD has been characterized in terms of output power, spectrum, beam profile, divergence angles and spatial coherence. Here, we provide measurements that were omitted from the main text for clarity.

In Fig. S2a, we plot the output power and the full width at half maximum (FWHM) linewidth as a function of current I. We obtain a threshold current I_{TH} = 90 mA, a

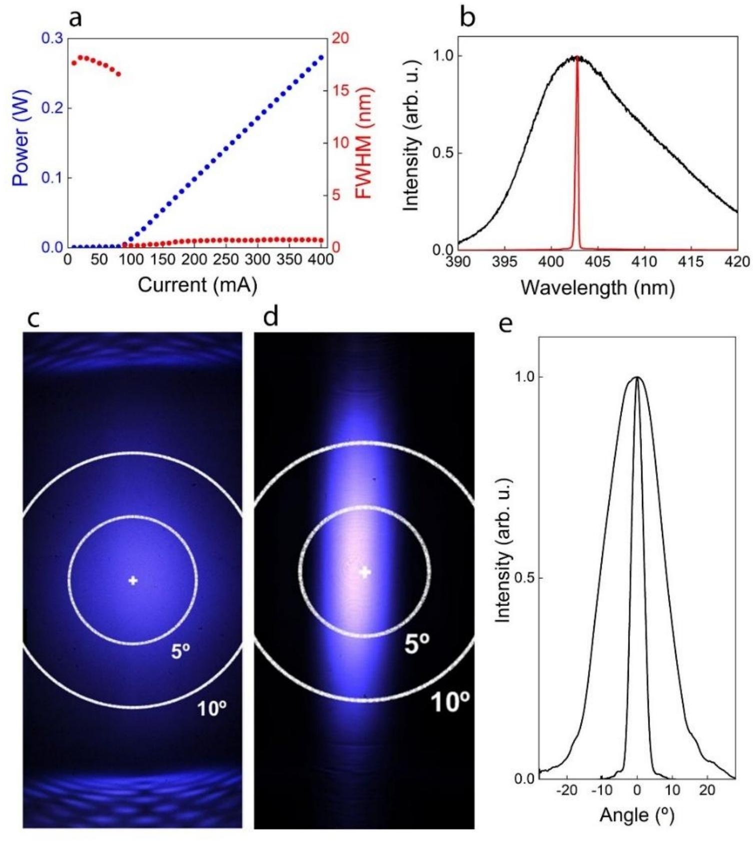

Fig. S2. (a) Output power (blue dots, left axis) and FWHM linewidth (red dots, right axis) as a function of current. (b) Spectra below (80 mA, black line) and above (120 mA, red line) threshold. Output beam below (c) and above (d) threshold, with I = 80 mA and I = 180 mA, respectively. (e) Cross sections of the output beam for I = 180 mA, along the horizontal (∥) and vertical (⊥) direction.

maximum power of 270 mW at 400 mA, and an efficiency slope η_{SLOPE} = 0.9 W/A. The FWHM linewidth between 18 nm and 16.5 nm, below I_{TH}, then narrows down to 0.2 nm right above threshold and slightly increases to 0.7 nm at high injection currents. Two typical spectra below (80 mA) and above (180 mA) I_{TH} are plotted in Fig. S2b.

Measurements of the output beam are performed with a CMOS camera (Imaging Source DFK 37BUX250, 3.45 μm pixel) by placing the camera in front of the LD at a distance of 2.8 cm. As the beam size is greater than the camera size (7.06 mm x 8.44 mm) we move the camera on the plane orthogonal to the propagation axis and stitch adjacent captures.

The output beam is measured below (c) and above (d) I_{TH}, with I = 80 mA and I = 180 mA, respectively. In Fig. S2c, we observe a central ellipsoid beam and two satellite emission regions, above and below the ellipsoid, which are attributed to parasitic substrate modes [1,2]. These features become negligible when the original LD is biased above threshold (Fig. S2d), however they do appear in modified devices, due to the modification of the waveguide form the ablation process, as discussed in Section 3. Cross sections of the output beam for I = 180 mA, along the horizontal (∥) and vertical (⊥) direction are shown in Fig. S2e. The divergence angles calculated at $1/e^2$ are of 6° and 32°, in the horizontal and vertical direction, respectively.

Section 2. Ablation Technique.

The ablation set-up is shown in Fig. S3a. A picosecond pulsed laser (PL, Ekspla PL2551C) is used. The collimated output beam (in light red) of the pulsed laser (PL)

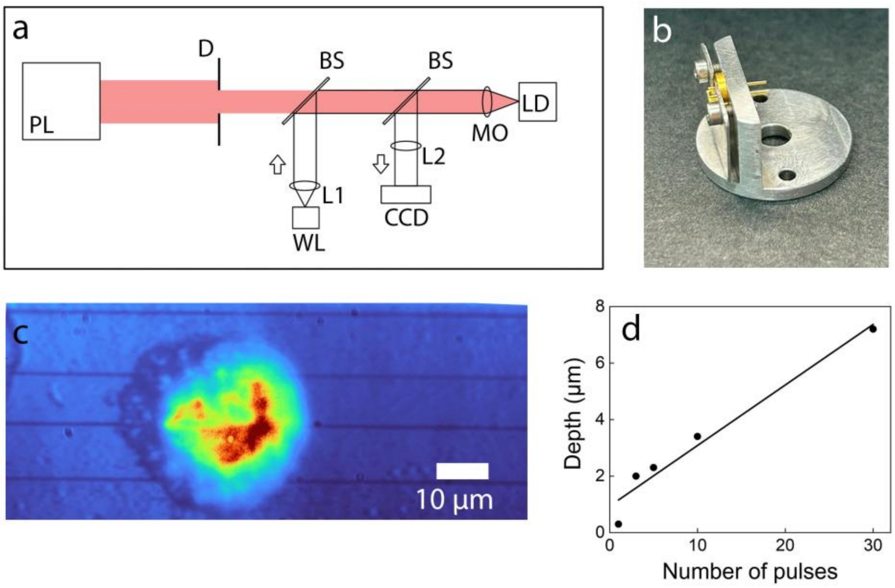

Fig. S3. (a) Ablation set-up, details are given in the text. (b) L-shaped adapter for LD mount. (c) Image of the ablated area in a sacrificial device, when biased with few milliamperes. (d) Measured depths (black dots) as a function of the number of delivered pulses and linear fit (black line).

has its diameter controlled with an iris diaphragm (D) and is focused with a microscope objective (MO, numerical aperture 0.55 and magnification x40) on the laser diode (LD). The ablation process is monitored in real time, illuminating the LD with a white light (WL) source and imaging the LD surface on a CCD camera. Lenses L1 (focal length of 3 cm) and L2 (focal length of 5 cm) are used for collimating the WL output and imaging onto the CCD, respectively. The pulse energy is measured using an energy meter positioned upstream of the microscope objective prior to the ablation process.

The LD is located on a laser mount (Thorlabs LDM56) and connected to current (Thorlabs LDC205C) and temperature (Thorlabs TED200C) controllers. The laser mount Thorlabs LDM56 is a standard LD mount intended to be used with edge emitting LDs. In our technique, the ablation process must access the top surface of an edge emitting LD. For this reason, we have built an ad hoc adapter shown in Fig. S3b. The L-shaped adapter allows the ablation beam to be focused on the top surface of the LD, maintaining thermal contact and electrical access.

Current injection allows biasing the device during the ablation process so that light emission can be seen from the perforated holes, for checking that the active layer has been reached. The ablation spot is positioned on the LD surface by acting on a 3-axis translation stage where the LD and its mount are placed. The process is observed in real time via the CCD camera. In Fig. S3c, we show the image detected at the CCD after the ablation process is completed. We observe four horizontal lines corresponding to the ridge waveguide seen from above and the light spot emerging from the ablated hole.

The selection of ablation parameters, i.e. pulse diameter, pulse energy and number of delivered pulses, is discussed in the main manuscript. We set a diameter of 10 μm and a single pulse energy of 80 μJ, corresponding to a single pulse flux of 100 J/cm^2. We delivered different numbers of pulses on the ridge waveguide of a sacrificial LD and we measured the obtained depth by means of optical profilometry. In Fig. S3d, we show the measured depths (black dots) as a function of the number of delivered pulses and the corresponding linear fit (black line), which gives an ablation depth of about 0.2 μm per pulse.

Section 3. Modified devices

Three devices have been fabricated. Their structures are schematically depicted in Fig. S4, where devices labelled A, B and C are seen from above and the three rectangular stripes represent the ridge waveguide structure as observed from above.

In each device, we introduce a scattering hole, obtained with the same ablation parameters (pulse energy, beam diameter and number of pulses), and positioned in three different locations: i) near the back mirror (device A, presented in previous version of the manuscript), ii) at mid-length of the cavity (device B) and iii) at the front

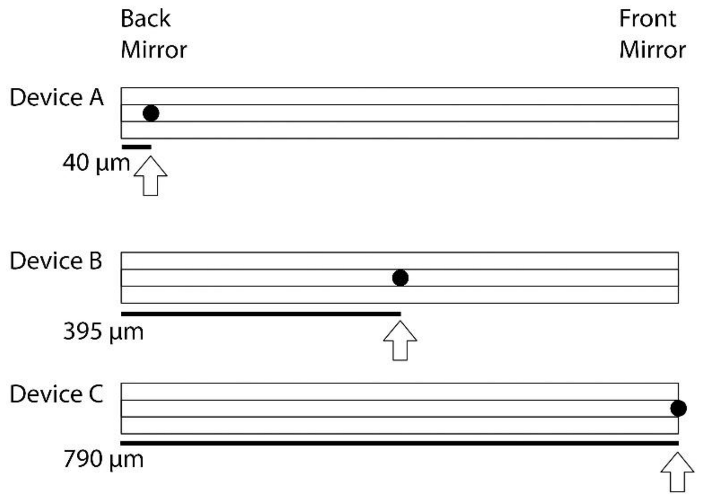

Fig. S4. Schematic representation of the three fabricated devices.

mirror location (device C). Specifically, the center of the scattering hole is located at 40 µm, 395 µm and 790 µm from the back mirror, for device A, B and C, respectively.

These three positions are selected because they break the waveguide in three representative points: in device A and C we maximize the amplification path length and obtain the output beam from the untouched mirror (device A) and from the induced rough surface (device C). In device B, amplification length is reduced to half the cavity, and the output is from the unmodified mirror. In this way, we aim to observe how the defects affect the emission properties depending on different amplification paths and output surfaces.

In device A, the scattering hole is located as close as possible to the back mirror, in fact the back mirror position is shadowed by the remaining part of the TO-can encapsulate (see Fig S1a) and the ablating beam cannot reach its position.

The three devices are characterized in terms of output power, spectrum, speckle contrast and beam shape. Results for device A are presented in the main manuscript, in this Section we present results for devices B and C.

3.1. Device B.

The output power and FWHM linewidth of device B are shown in Fig. S5a as a function of current. We observe an increase in output power up to a maximum of 0.24 mW and a decrease in linewidth from 10.7 nm to 7.6 nm. In Fig. S5b, we plot the speckle contrast versus current, which increases from 0.1 to 0.16.

In Fig. S5c, we show the beam profile measured with a CMOS camera (Imaging Source DFK 37BUX250, 3.45 µm pixel) by placing the camera in front of the LD at a distance of 2.8 cm. As the beam size is greater than the camera size (7.06 mm x 8.44 mm), we move the camera on the plane orthogonal to the propagation axis and stitch adjacent captures.

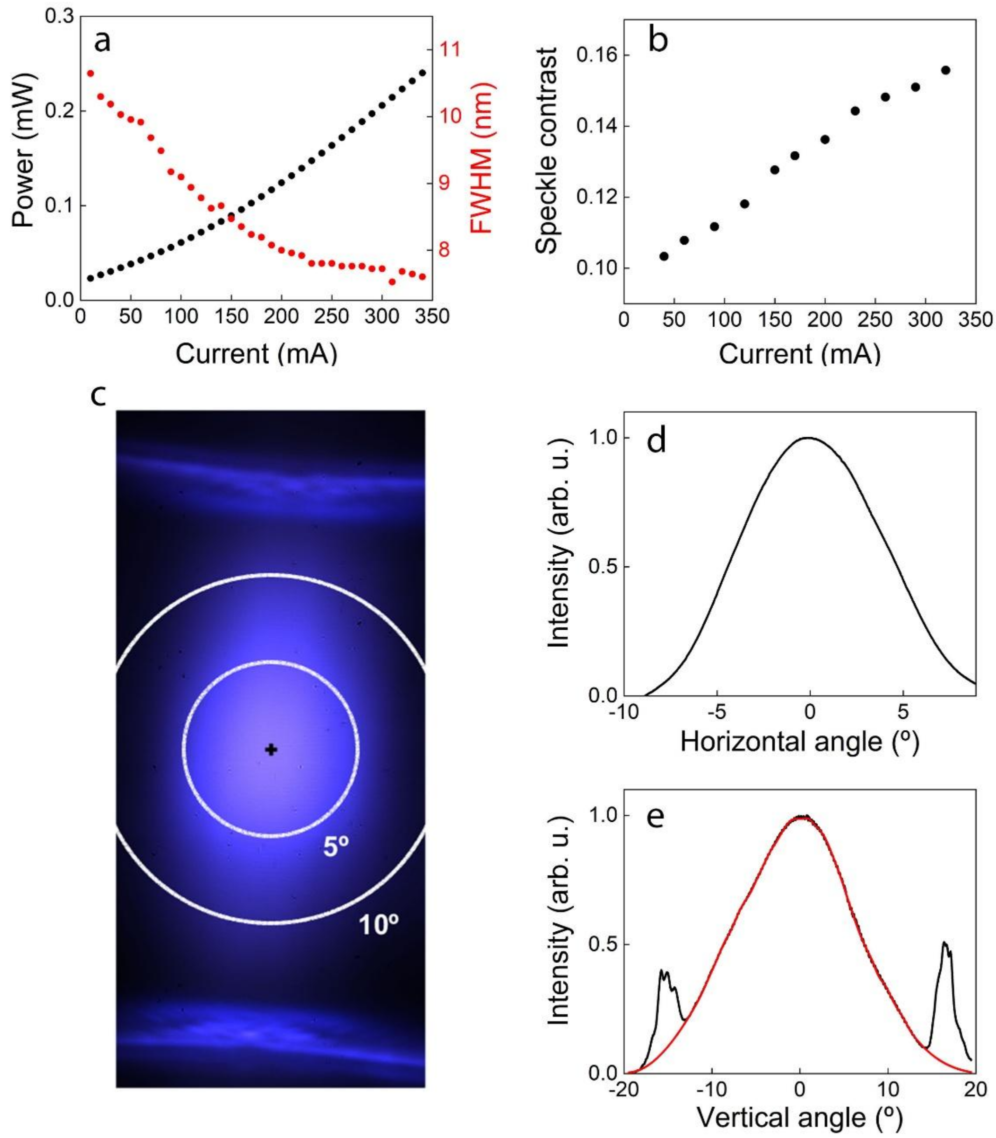

Fig. S5. Emission characteristics of device B: a) output power (black dots, left axis) and spectral linewidth (red dots, right axis) versus current; b) speckle contrast versus current; c) beam profile at 300 mA; d) horizontal and e) cross section of the beam. Details in the text.

The output beam consists of a central ellipsoid and two curved stripes above and below it, along the vertical direction. These are attributed to unwanted light propagation into the substrate, as observed in previous studies on ridge waveguide gallium nitride LDs [1,2].

In those works, the LD emission is studied considering the effect of the index gap between the waveguide and the cladding layers. Poor vertical confinement results in satellite lobes along the vertical direction of the near field (Fig. 6 in [1]). In the original LDs of our study, substrate modes are also observed, although above threshold, their intensities are negligible with respect to the central part of the beam. In our modified devices, the vertical confinement deteriorated due to the ablation process which perforates both the cladding layers and the waveguide, resulting in strong propagation into the substrate and producing the two horizontal stripes observed in Fig. S5c.

In Fig. S5d, we plot the horizontal cross section of the beam, which has a divergence angle at $1/e^2$ of 14º.

In Fig. S5e, we plot the vertical cross section of the beam, with the two side lobes corresponding to substrate modes. We obtain a divergence angle of 35º considering the measured curve and a divergence angle of 27º after interpolation of the central part of the curve (red line in Fig. S5e) with a modified Akima method.

3.2. Device C.

Results from device C are shown in Fig. S6. In Fig. S6a, we plot the output power and FWHM linewidth as a function of current. We observe a maximum output power of 2.7 mW and a linewidth decrease from 12.6 nm to 4.8 nm. The speckle contrast increases as a function of current from 0.15 to 0.2 (see Fig. S6b).

The beam is shown in Fig. S6c. We observe a region of the beam consisting of vertical lobes of intensity positioned slightly above the central direction of propagation (red rectangle). This part of the beam is not centered and is not an ellipsoid as in device A and B. In Fig. S6d, we sum the content of the red rectangle along the vertical direction, obtaining the horizontal profile of this part of the beam. The corresponding divergence angle at $1/e^2$ in the horizontal direction is 25º.

In Fig. S6c, the curved stripes due to emission from substrate modes are more intense than the central part of the beam. This is also observed in Fig. S6e, where we sum the full image of Fig. S6c along the horizontal direction, obtaining the vertical profile of the entire emission. The higher peak at negative angles corresponds to the horizontal, intense stripe of Fig. S6c located below the center of propagation, i.e. towards the substrate. The other peak at positive angles corresponds to the stripe of Fig. S6c located above the center of propagation.

From Fig. S6e, we obtain a divergence angle at $1/e^2$ of 38º considering the measured curve and a divergence angle of 32º after interpolation of the central peak (red dots in Fig. S6e) with a modified Akima method.

Device	Max output (mW)	Min linewidth (nm)	Max contrast	div angles (hor)	div angles (ver)
A	1.8	5.9	0.2	12	35
B	0.24	7.5	0.16	14	35
C	2.8	5	0.2	25	38

Table S1. Comparison of emission characteristics of devices A, B and C.

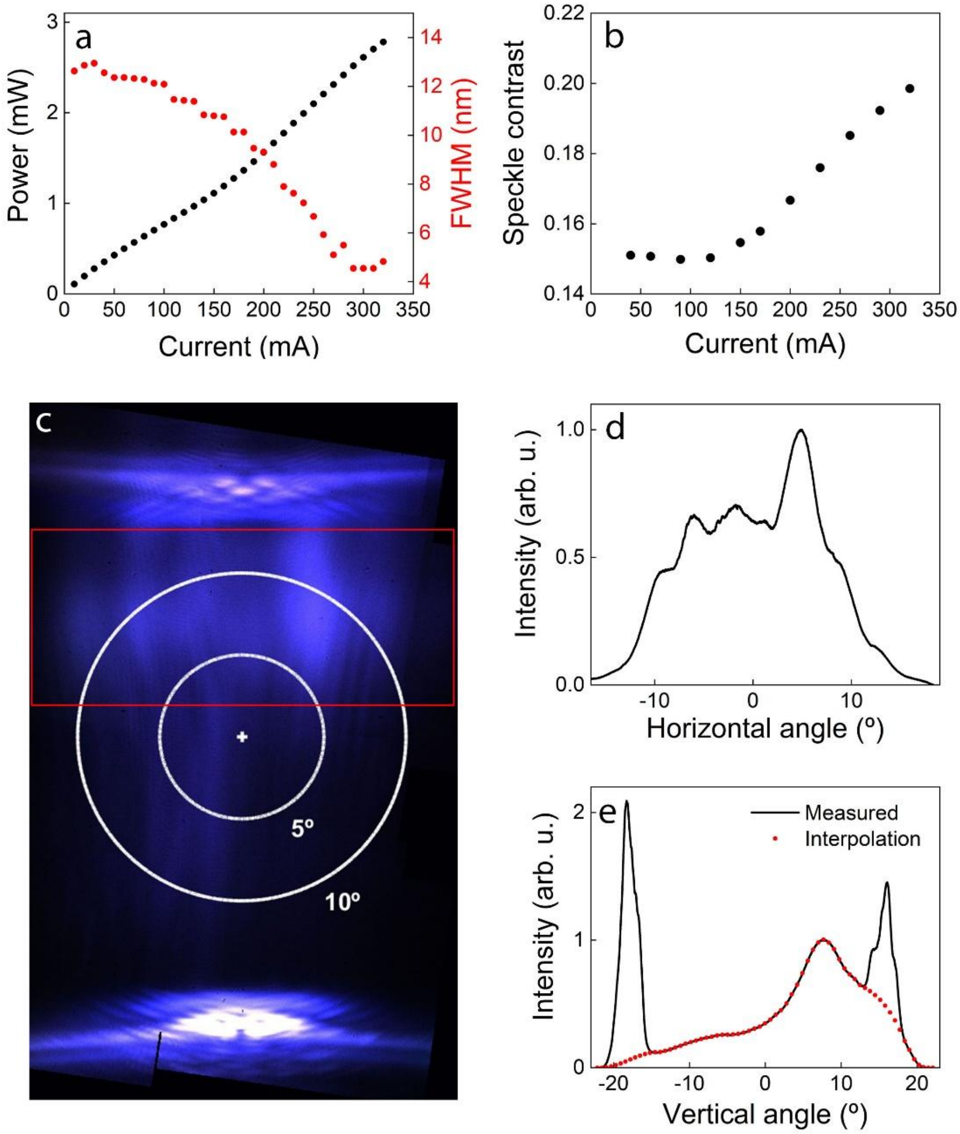

Fig. S6. Emission characteristics of device C: a) output power (black dots, left axis) and spectral linewidth (red dots, right axis) versus current; b) speckle contrast versus current; c) beam profile at 300 mA; d) horizontal and e) vertical cross section of the beam. Details in the text.

Section 4. Numerical simulations.

We model the emitted power from our device as [3–5]:

$$P = P_0 \cdot \exp\left[(\Gamma \cdot g(I) - \alpha) \cdot L\right] \quad (1)$$

where P_0 is a scaling factor, Γ is the confinement factor, g(I) is the current dependent gain, α the total losses and L the amplification path length. The current dependent gain is defined as:

$$g(I) = G_P[1 - \exp(-g_0 \cdot I)] \quad (2)$$

where G_P is the maximum gain (cm^{-1}), g_0 is the current conversion parameter (A^{-1}) and I is the current.

We fix the parameter G_P = 4000 cm^{-1} [6], because G_P and Γ only appear through their product, making them non-identifiable when varied simultaneously. We let the other parameters, i.e. P_0, Γ, g_0 and α, free to vary.

The fitting procedure is based on the least square fitting method, and we provide as inputs the injection current and the measured output power. The relative error is estimated by calculating the root mean square of the relative residuals (RMS_{REL}) as:

$$RR_{RMS} = \sqrt{N^{-1} \cdot \sum_{i=1}^{N} r_i^2}$$

where the relative residuals are defined as $r_i = (m_i - f_i)/m_i$, with m_i and f_i, the measured and fitted values, respectively, and N the total number of points.

The measured (black dots, left axis) and fitted (black crosses, left axis) output power and calculated gain g(I) (red dots, right axis) versus current for devices B and C, in Fig. S7a and S7b, respectively. Results for device A are plotted in Fig. 2 of the main manuscript.

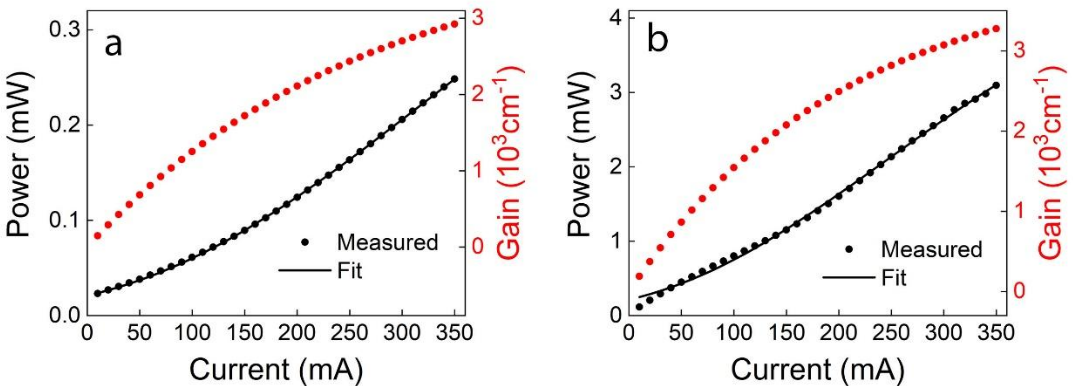

Fig. S7. Measured (black dots, left axis) and fitted (black crosses, left axis) output power and calculated gain g(I) (red dots, right axis) versus current for devices B (a) and C (b).

In Tab. S2, we present extracted parameters for each device and the RMS_{REL} for each simulation. Parameters obtained from simulations should be considered with caution, knowing that having four free parameters allow considerable flexibility and different sets of parameters can produce equally plausible fits.

In the main manuscript we compare the losses obtained for device A with the original LD. Simulation results from device C present an RMS error = 21%, possibly due to the parasitic modes from the substrate. We limit our qualitative comments to parameters extracted from device A and B.

From device B, we obtain higher losses compared to device A, which reflects the considerably lower output and lower parameter g_0, which accounts for a less saturated gain curve, as expected in a shorter device. Confinement factor is similar in both devices.

	P_0	Γ (%)	g_0 (mA^{-1})	α (cm^{-1})	RMS_{REL}
Device A	19	1.9	4.9	93	6%
Device B	11	2.1	3.8	159	1%
Device C	36	1.0	4.9	65	21%

Tab. S2. Extracted parameters for each device and RMS_{REL} of fitted curved.

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